DEEP LEARNING CIE SUBMISSION

PBL TOPIC: ATTENTION-BASED TEXT CLASSIFICATION

Student Name:	Student 4 (Group Member 4)
Roll No:	34134
Project Topic:	Attention-Based Text Classification
Paper Reviewed:	*Hierarchical Attention Networks for Document Classification*
Authors & Year:	Zichao Yang, Diyi Yang, Chris Dyer, Xiaodong He, Alex Smola, Eduard Hovy (2016)
Publication Venue:	NAACL-HLT 2016 | ACL Anthology N16-1174

1. Problem Addressed

Document classification requires capturing hierarchical structures: words form sentences, and sentences form documents.

2. Motivation & Need

Not all words and not all sentences are equally informative for document sentiment classification.

3. Methodology & Model Architecture

Proposed Hierarchical Attention Networks (HAN): Word-level Attention extracts sentence vectors, and Sentence-level Attention extracts document vectors.

4. Dataset & Experimental Setup

Yelp Reviews, IMDb Movie Reviews, Amazon Customer Reviews, and Yahoo Answers datasets.

5. Attention Mechanism Formulation

Two-level additive attention mechanism with learnable word context vector u_w and sentence context vector u_s.

6. Key Results Reported in Paper

Outperformed standard LSTMs, GRUs, and CNNs across all six document classification benchmark datasets.

7. System Advantages & Strengths

Hierarchical structure mirrors natural language hierarchy and yields highly interpretable word and sentence heatmaps.

8. Limitations & Constraints

Requires sentence boundary segmentation preprocessing and increases parameter count.

9. Identified Research Gap

Flat sequence models failed to model the hierarchical word-sentence structure of documents.

10. Relevance to Our PBL Project

Directly inspires our token-level attention heatmap visualization for IMDb movie reviews.

11. Student's Critical Observations

Demonstrated that attention visualization provides actionable insights into document sentiment classification.

12. References

[1] Z. Yang et al., 'Hierarchical attention networks for document classification,' in NAACL-HLT, 2016.